NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-20/29C
9 May 1989
SUPERSEDING
MIL-C-20/29B
26 March 1984

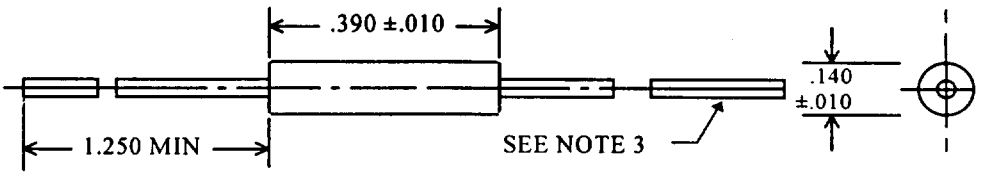
PERFORMANCE SPECIFICATION SHEET

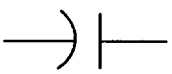
CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR77 AND CC77

STYLE CC77 IS INACTIVE FOR DESIGN AFTER 26 March 1984. USE CCR77.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-20.





CIRCUIT DIAGRAM

Inches	mm
.001	0.03
.005	0.13
.010	0.25
.016	.041
.090	2.29
.160	4.06
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .015 (0.38 mm) to .027 (0.69 mm).
© 4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Styles CCR76 and CC76 capacitors.

© denotes changes

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR77 only): M (1.0%), P (0.1%), R (.01%) or S (.001%).

Thermal shock and voltage conditioning (CCR77 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

TABLE I. Capacitor characteristics.

Type 1/ 2/ designation	Rated voltage	Nom. cap.	Cap. tol.	Type 1/ 2/ designation	Rated voltage	Nom. cap.	Cap. tol.
	Volts, dc	pF			Volts, dc	pF	
CC-77CG151--	200	150	FGJ	CC-77CG102--	100	1000	FGJ
CC-77CG161--	200	160	FGJ	CC-77CG112--	100	1100	FGJ
CC-77CG181--	200	180	FGJ	CC-77CG122--	100	1200	FGJ
CC-77CG201--	200	200	FGJ	CC-77CG132--	100	1300	FGJ
CC-77CG221--	200	220	FGJ	CC-77CG152--	100	1500	FGJ
CC-77CG241--	200	240	FGJ	CC-77CG162--	100	1600	FGJ
CC-77CG271--	200	270	FGJ	CC-77CG182--	100	1800	FGJ
CC-77CG301--	200	300	FGJ	CC-77CG202--	100	2000	FGJ
CC-77CG331--	200	330	FGJ	CC-77CG222--	100	2200	FGJ
CC-77CG361--	200	360	FGJ	CC-77CG242--	50	2400	FGJ
CC-77CG391--	200	390	FGJ	CC-77CG272--	50	2700	FGJ
CC-77CG431--	200	430	FGJ	CC-77CG302--	50	3000	FGJ
CC-77CG471--	200	470	FGJ	CC-77CG332--	50	3300	FGJ
CC-77CG511--	200	510	FGJ	CC-77CG362--	50	3600	FGJ
CC-77CG561--	200	560	FGJ	CC-77CG392--	50	3900	FGJ
CC-77CG621--	200	620	FGJ	CC-77CG432--	50	4300	FGJ
CC-77CG681--	200	680	FGJ	CC-77CG472--	50	4700	FGJ
CC-77CG751--	100	750	FGJ	CC-77CG512--	50	5100	FGJK
CC-77CG821--	100	820	FGJ	CC-77CG562--	50	5600	FGJK
CC-77CG911--	100	910	FGJ				

1/ Complete type designation will include the following:
1st dash - Symbol "R" (for style CCR77) or dash will be deleted (for style CC77).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR77 only) or dash will be deleted (for style CC77).

2/ Capacitors deleted since the previous issue shall be replaced by selecting a capacitor with the nearest parameter.

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Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: Method I of MIL-STD-1285, as shown in following example:

	ER	
	Type designation	CCR77C G151FM
©	"J" brand and CAGE code	J12345
	Date code and lot code (lot code may require two characters)	8942A

	NON-ER	
	Type designation	CC77C G151F
©	CAGE code	12345
	Date code and lot code (lot code may require two characters)	8942A

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
DLA - ES

User activity:
Navy - AS, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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